

FEATURE GUIDE

PineTime + Companion

Everything the watch and its companion app can do —
feature by feature, on both screens.



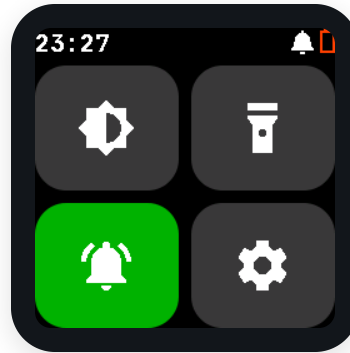
InfiniTime firmware (fork) · PineTime Companion app · generated from live device & simulator screenshots

The watch at a glance

A clean digital watch face, with every app one swipe away. The launcher holds sixteen apps across three pages — fitness, tools, games, and the custom additions below.

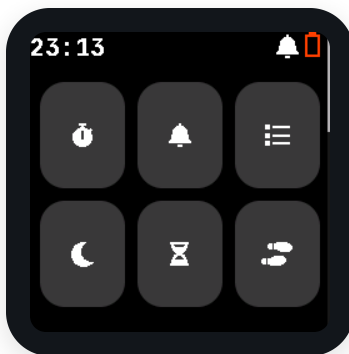


Watch face

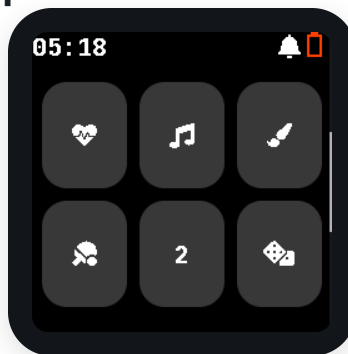


Quick settings (swipe right)

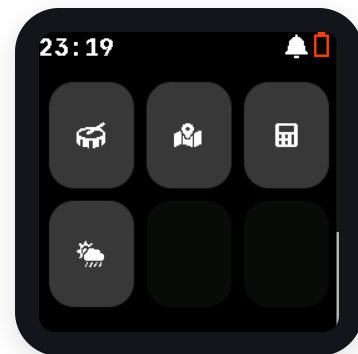
App launcher — all sixteen apps



Page 1



Page 2



Page 3

FLAGSHIP FEATURES

What makes this build special

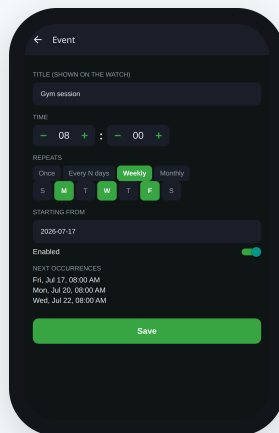
Capabilities the companion drives end-to-end — shown on the watch and in the app.

Schedule & reminders

Recurring reminders that live on the watch



On-watch schedule list



Companion → a complex recurring event

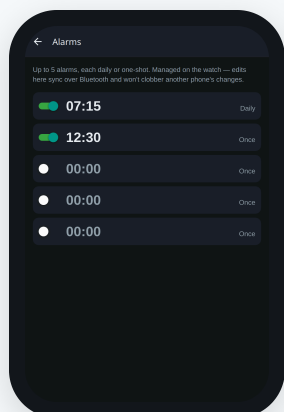
- Build complex recurrences: every N days, weekly on chosen weekdays (e.g. Mon/Wed/Fri), or a day of the month — with a live preview of the next occurrences.
- Reminders sync to the watch and fire on-wrist even when the phone is away; the watch shows them in a scrollable list.
- Multi-phone-safe sync: edits from a second phone merge instead of clobbering.

Alarms

Up to five daily or one-shot alarms



Multi-alarm app on the watch

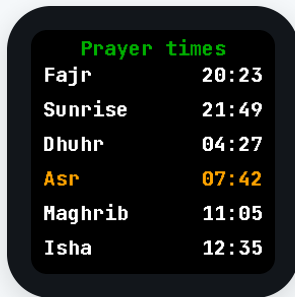


Companion → Alarms

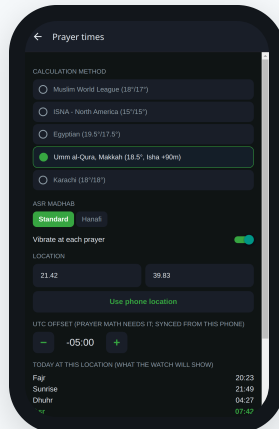
- Manage up to five alarms, each daily or one-shot, from the phone — or on the watch itself.
- Changes sync both ways with conflict-safe versioning, so on-watch edits are preserved.
- A custom InfiniTime service replaces the stock single-alarm app.

Prayer times

Five daily prayers, computed on-device



Prayer-times app on the watch



Companion → Prayer times

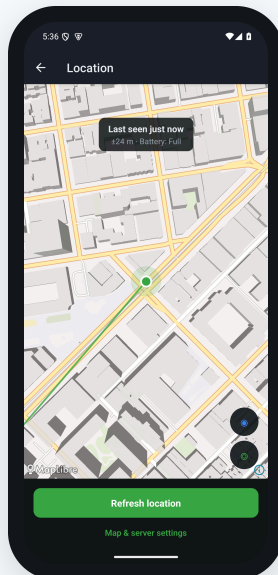
- Pick a calculation method (Muslim World League, ISNA, Egyptian, Umm al-Qura, Karachi...) and Asr madhab.
- Set the location from the phone's GPS or enter coordinates; the watch computes the five daily times itself and shows them in its own app.
- Optional vibration at each prayer time.

Find My (locator beacon)

Turn the watch into an Apple Find My tag



Find My toggle on the watch



Companion → location history on a map

- Generate a key pair, provision it to the watch, and turn beaconing on (Settings → Find My); nearby iPhones then report its location to Apple's Find My network.
- The app pulls the crowd-sourced fixes and plots the watch's location — with its recent trail and accuracy — on a live map.
- Export the keys to also look the watch up in your own macless-haystack server.

Weather

Live forecast pushed to the wrist



Weather app → current + 5-day forecast



Companion → Weather

- Fetched from Open-Meteo (no account, no API key) for the watch's location — its prayer-times coordinates or the phone's GPS.
- Current temperature, icon, and a five-day forecast reach the Weather app and the watch-face widget, refreshed on every connect.
- The watch keeps it for 24 hours and converts °C/°F itself — real data on the wrist even when the phone is away.

Step history

Durable daily step tracking on the phone



Steps app on the watch



Companion → 14-day step chart

- The watch counts steps itself but only holds today and yesterday; the app reads and keeps the durable daily history.
- A 14-day bar chart (today highlighted, goal line) shows the trend — tap any bar for that day's count.
- Reading the watch's yesterday total backfills the previous day, so a late or missed read never undercounts.

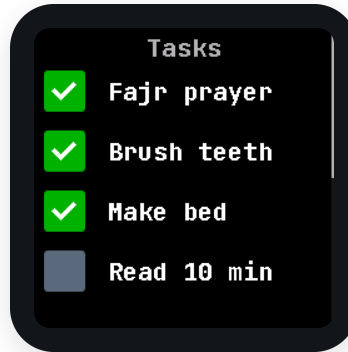
FLAGSHIP FEATURES

Daily tasks — a checklist that resets every day

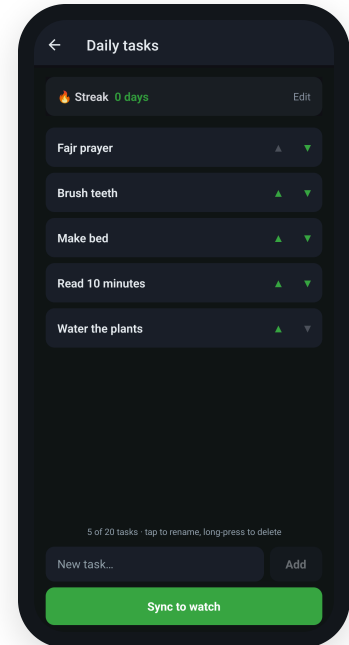
A fixed daily routine you build on the phone and tick off on the wrist. The watch shows how many are done today with a progress ring and a streak; everything un-checks at midnight.



Watch → today's progress + streak



Watch → tap a row to tick it off



Companion → edit the daily list

Built on the phone, ticked on the wrist. Add, rename, reorder and delete tasks from the companion (up to 20). The list syncs to the watch, where the wearer only ticks them off — no fiddly editing on the tiny screen. Ticks live **only on the watch** and clear at local midnight, so every day starts fresh.

Streaks keep it going. Finish every task in a day and the watch bumps a streak counter at midnight; miss one and it resets to zero. The streak shows on the watch summary, and the phone can read it or set it directly — forgive a missed day, or hand out a reward. Like the schedule, the list is watch-authoritative: edits from a second phone merge instead of clobbering.

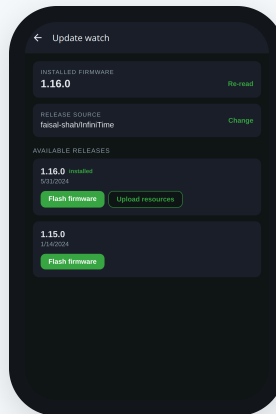
Firmware & resources over the air

Firmware & resources update (OTA)

Update InfiniTime straight from the app



“Firmware & files” must be enabled



Companion → Update watch

- Lists releases from a configurable GitHub repo, reads the installed version, and flashes firmware over Nordic Legacy DFU — no nRF Connect needed.
- Pushes the matching external-resources pack (fonts/images) over the BLE filesystem.
- After a flash it reminds you to tap Validate on the watch, then re-checks the version to confirm it stuck.

How an update runs. Pick a release → the app downloads it → streams the firmware to the watch in the mandatory 20-byte DFU packets → the watch reboots into the new image **unvalidated**. Back and swipe navigation stay locked until the transfer ends, and duplicate starts are rejected; keep the app in the foreground. You then tap **Settings** → **Firmware** → **Validate** on the watch to keep it — skip that and the next reboot rolls back. Finally the app pushes the matching resources over the BLE filesystem.

Where it works. Firmware DFU runs on **Android** and against the simulator; a plain web browser can't reach the DFU service (it's on Chromium's Bluetooth blocklist), so the app hides firmware there. Resource uploads work everywhere. Both require “**Firmware & files**” enabled on the watch.

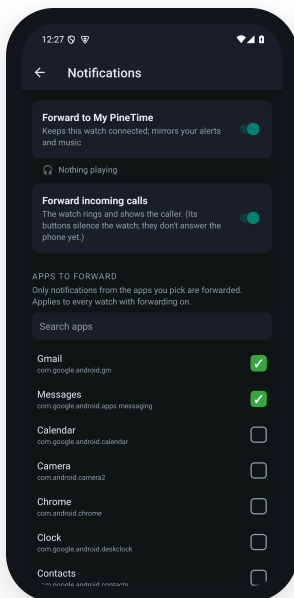
NOTIFICATIONS & ALERTS

Your phone on your wrist

On Android the app keeps a background connection to the watch and mirrors your phone — notifications, incoming calls, and now-playing music — all under one per-watch switch.



Forwarded notification



Companion → Notifications



Now-playing on the Music app

Notification forwarding, per watch. Turn on **Forward notifications** for a watch and the alerts from the apps you choose (a searchable allowlist) show up on its wrist; incoming calls ring it with the caller's name. It's **per watch on purpose** — forward to your own watch but not the kids'. A grant of Android's Notification Access is all it needs; a foreground service keeps the link alive with the app swiped away.

Granting Notification Access on Android 13+. Android hides this permission for any app installed outside the Play Store, and the greyed-out switch just says “Restricted setting”. The unlock sits in an overflow menu: tap the greyed-out switch and press OK, go to **App info**, tap **⋮** → **Allow restricted settings**, then come back and grant access. The app's Notifications screen deep-links to both pages and spells out these steps if the first attempt doesn't take.

What each switch covers. **Forward incoming calls** works on its own — no phone app needs picking, because a ringing call is flagged as a call rather than an ordinary notification. Missed calls and voicemail are ordinary notifications, so tick your dialer in the app list if you want those too. **Now playing** and the watch's music controls need only Notification Access; the app list doesn't affect them.

Music control rides the same switch. The watch's Music app shows the current track, artist and progress from whatever's playing on the phone, and its buttons drive the phone back — play/pause, next/previous, and volume. Change the song on the phone and the wrist follows; skip on the wrist and the phone skips.

NOTIFICATIONS & ALERTS

Watch alerts that don't get lost

Separately from forwarded phone notifications, alerts the watch raises itself — alarms, schedule reminders, prayer times — share one queue so a new one never tramples the last.



The pending-alerts queue (1 of 2)



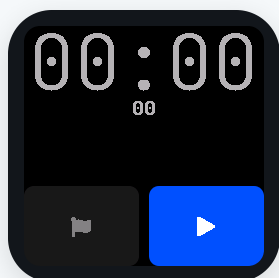
Reminders live on the watch

The pending-alerts queue. Every watch-originated alert — a multi-alarm going off, a schedule reminder, a prayer-time alert — lands in one shared queue instead of each app owning a screen that the next alert would trample. Miss one while another is ringing and it still waits for you: the queue holds the most recent alerts (newest first), shows a **1 / N** counter so you can page through them, and you clear each with **OK**.

BUILT-IN APPS

Everyday apps on the watch

Standard InfiniTime apps, all running on-wrist. Four more — Paint, Paddle, 2048 and Dice — round out the games (see launcher page 2).



Stopwatch

Start / lap / reset timing.



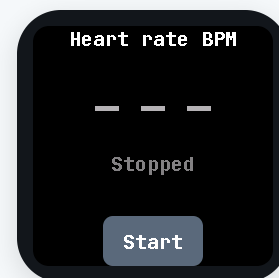
Timer

Countdown with vibration alert.



Steps

Daily count + on-phone history.



Heart rate

On-demand optical HR reading.



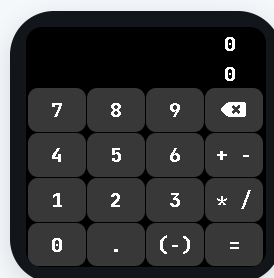
Music

Controls the phone's playback.



Navigation

Turn-by-turn arrows from the phone.



Calculator

Basic on-wrist calculator.



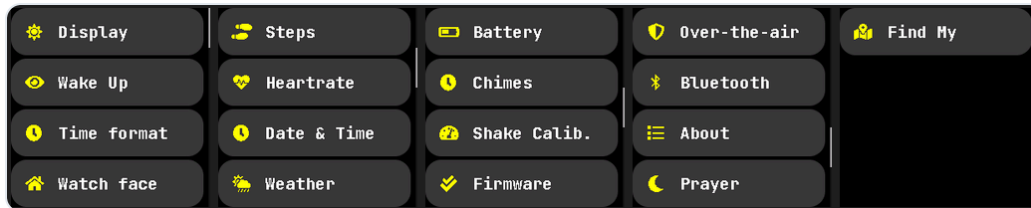
Metronome

Adjustable tempo with haptic beat.

SETTINGS & CONNECTIVITY

Settings and Bluetooth

The on-watch settings cover the display, watch face, sensors, and connectivity. Two settings gate everything the companion does.



The full settings list (five pages)



Bluetooth: radio on/off



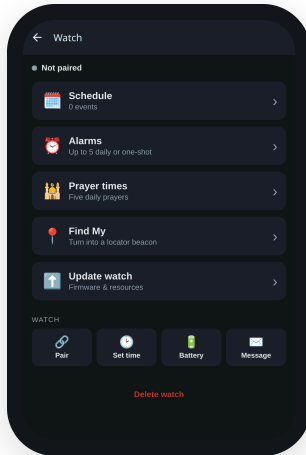
“Firmware & files”: OTA gate

How they relate. **Bluetooth** is the master radio switch — turning it off stops all connections (and force-stops the Find My beacon). **Firmware & files** must be enabled for firmware DFU and resource uploads; if it's off the watch refuses them and the app tells you to turn it on. The Find My beacon needs Bluetooth on and a provisioned key, and while it broadcasts it takes over advertising, so the watch isn't pairable at the same time.

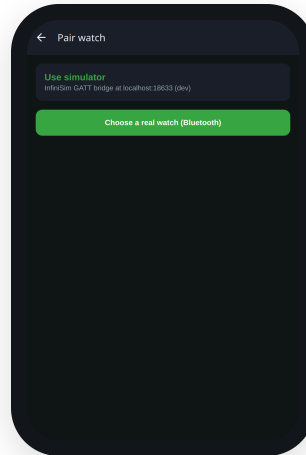
COMPANION APP

Managing watches from the app

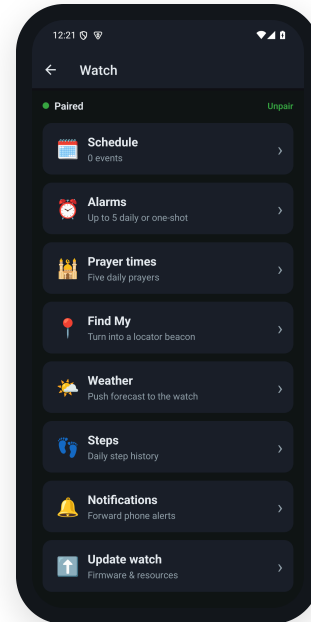
The app runs on Android, and on web/desktop for everything except firmware DFU. Add a watch, pair it, and each watch gets its own hub of features.



Your watches



Pair — real watch or simulator



The watch hub

From the hub you reach every feature — Schedule, Alarms, Prayer times, Find My, Weather, Steps, Notifications, and Update — plus the quick actions: **Set time**, read **Battery**, and send a **Message** to the watch. Pairing works against a real watch over Bluetooth or against the InfiniTime simulator during development.

Verified pairing, and honest repair. A watch remembers up to **five companion phones**; pairing a sixth evicts the least-recently-used one after confirmation. The app saves a watch only after encrypted verification. **Remove from app** leaves the system and watch bonds intact, while **Repair pairing** distinguishes a reset, LRU eviction, and an out-of-sync bond before guiding a system-Bluetooth re-pair. InfiniTime 2.0.2 imports no earlier bond format, so each companion pairs once after upgrading while watch data remains intact. Notification forwarding owns the watch's single link; the app reports that busy state instead of recommending needless re-pairing.